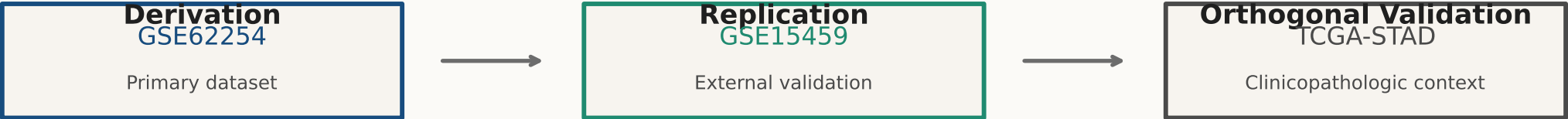


Study design, cohort qualification, and technical consistency

Frozen centroids, explicit cohort roles, and projection-confidence summaries define the validation engine.

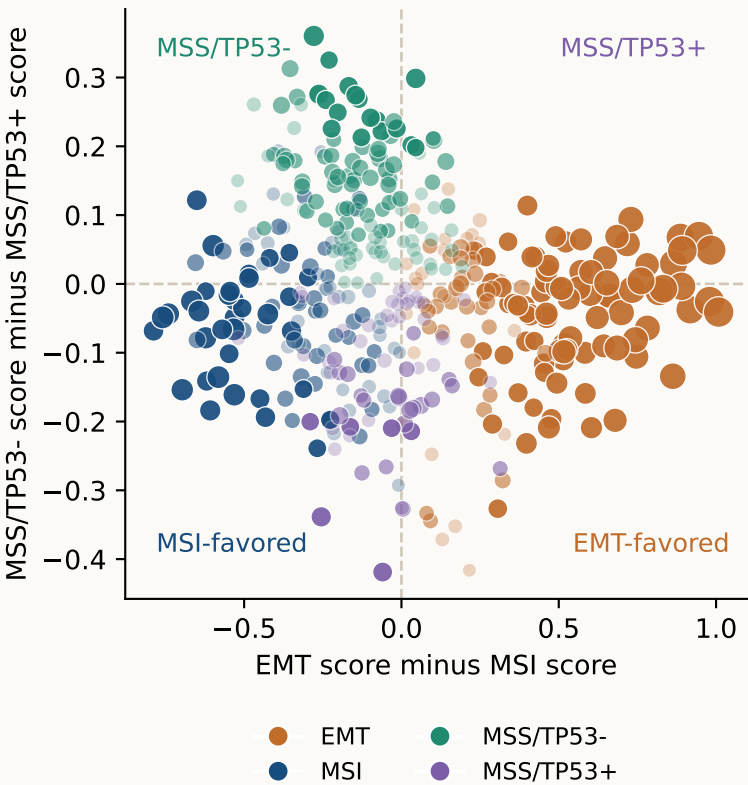
A Frozen derivation-to-validation workflow



B Cohort roles and endpoint context

Cohort	Role	N
GSE62254	Derivation	300
GSE15459	Replication	192
GSE84437	Replication	433 used
TCGA-STAD	Orthogonal	443
GSE26899	Tx extension	93
GSE26901	Tx extension	109

C Centroid space



D Projection QC

